Solving Polynomial Equations Modulo a Prime

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1. Let p be a prime number.
   (a) Find all solutions to $x^2 \equiv 1 \pmod{p}$.
   (b) For which primes p does $x^2 \equiv -1 \pmod{p}$ have a solution?
   (c) How many solutions does $x^2 \equiv a \pmod{p}$ have for a given $a \not\equiv 0 \pmod{p}$?
2. Find all solutions to $x^3 + x + 1 \equiv 0 \pmod{5}$.
3. Let $f(x) \in \mathbb{Z}[x]$ be a polynomial of degree d. Show that $f(x) \equiv 0 \pmod{p}$ has at most d solutions modulo p.